

1. Client Requirements & Scenario Overview

- **Project Title:** Ratlou Local Municipality Offices (Setlagole) Network Infrastructure
- **Client Profile:** Municipal Services / Local Government Office (Setlagole)
- **Scope of Work:** Design and deploy a segmented, secure, and scalable local area network for administrative staff, public wireless access, and future infrastructure expansion using an assigned /24 root addressing block.

2. Network Design & Architecture

- **Hierarchical Topology Model:** Core/Distribution layer handled by SW1-Core connected directly to edge access switches (SW2-Admin, SW3-Public, and SW4-Floor2-CR2).
- **Router-on-a-Stick Inter-VLAN Routing:** Router R1-Ratlou uses sub-interfaces over a single 802.1Q trunk link to facilitate inter-VLAN routing and enforce traffic segmentation.
- **Security & Isolation:** Administrative traffic (VLAN 10) is logically separated from public Wi-Fi traffic (VLAN 20) and management traffic (VLAN 99). Wireless access is restricted exclusively to public zones and authenticated expansion areas.

3. IP Addressing Scheme (VLSM Allocation Table)

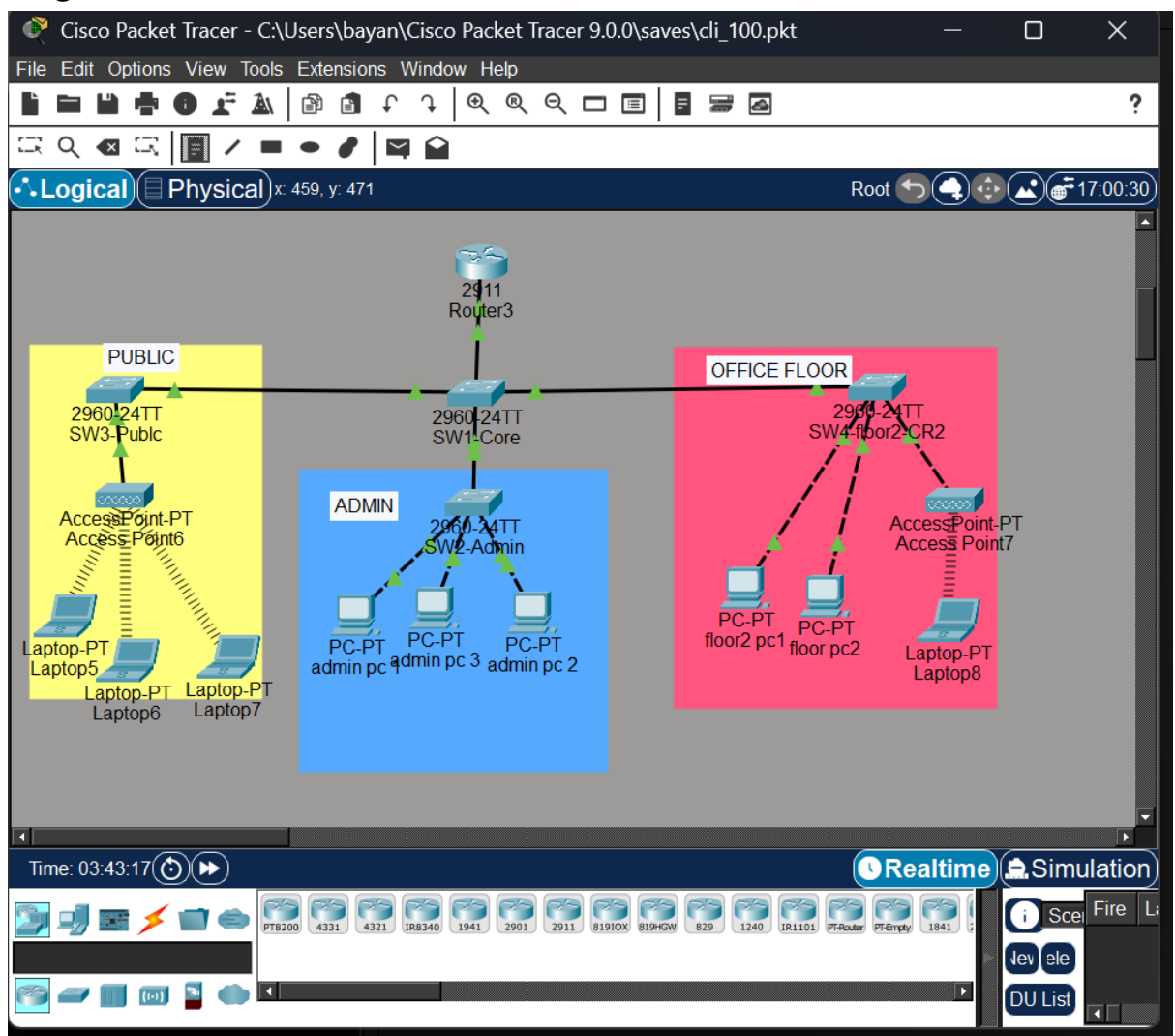
Root Network Block: 192.168.44.0/24

Subnet / Function	VLAN ID	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Admin & Staff	VLAN 10	192.168.44.0/26	255.255.255.192	192.168.44.1 - 192.168.44.62	192.168.44.1
Public Wi-Fi Zone	VLAN 20	192.168.44.64/27	255.255.255.224	192.168.44.65 - 192.168.44.94	192.168.44.65
CR2 Expansion Floor	VLAN 30	192.168.44.96/27	255.255.255.224	192.168.44.97 - 192.168.44.126	192.168.44.97

Subnet / Function	VLAN ID	Network Address	Subnet Mask	Usable Host Range	Default Gateway
Management Subnet	VLAN 99	192.168.44.128 /28	255.255.255.240	192.168.44.129 - 192.	192.168.44.129

4. Topology Layout

- Diagram Overview:**



- Device Inventory:**

- **Router:** R1-Ratlou (Cisco 2911 / ISR series)
- **Core Switch:** SW1-Core (Cisco Catalyst 2960)
- **Access Switches:** SW2-Admin, SW3-Public, SW4-Floor2-CR2 (Cisco Catalyst 2960)

- **Wireless Access Points:** Public AP (AP-PT) and Floor 2 WAP (AP-Floor2)
- **End Devices:** Admin PCs, Wireless Laptops, and Public PCs/Tablets

5. Packet Tracer Build Status

- Base network setup complete and functional.
- Inter-VLAN routing (802.1Q encapsulation) configured on router sub-interfaces.
- Active DHCP service running on R1-Ratlou for automated IP assignment across all subnets.
- Trunk ports configured across switches with native VLAN 99 alignment.

6. Configuration Commands (CLI Verification Logs)

code blocks pulled directly from **show running-config** in Packet Tracer:

1. Router (R1-Ratlou)

sub-interfaces, 802.1Q encapsulation, IP address assignments, and DHCP pool configurations:

! Router Sub-interfaces & Gateway Addresses

```
interface GigabitEthernet0/0.10
description Admin-Staff-Gateway
encapsulation dot1Q 10
ip address 192.168.44.1 255.255.255.192
```

!

```
interface GigabitEthernet0/0.20
description Public-Area-Gateway
encapsulation dot1Q 20
ip address 192.168.44.65 255.255.255.224
```

!

```
interface GigabitEthernet0/0.30
description CR2-New-Floor-Gateway
encapsulation dot1Q 30
ip address 192.168.44.97 255.255.255.224
```

!

```
interface GigabitEthernet0/0.99
description Management-Gateway
encapsulation dot1Q 99 native
ip address 192.168.44.129 255.255.255.240
```

! DHCP Pools Configuration

```
ip dhcp pool AdminPool
network 192.168.44.0 255.255.255.192
default-router 192.168.44.1
dns-server 8.8.8.8
```

!

```
ip dhcp pool PublicPool
network 192.168.44.64 255.255.255.224
default-router 192.168.44.65
dns-server 8.8.8.8
```

!

```
ip dhcp pool CR2Pool
network 192.168.44.96 255.255.255.224
default-router 192.168.44.97
dns-server 8.8.8.8
```

2. Core Switch (SW1-Core)

Show VLAN creations and all trunk interface settings:

! VLAN Database

```
vlan 10,20,30,99
```

! Router Trunk Link

```
interface GigabitEthernet0/1  
switchport mode trunk  
switchport trunk native vlan 99
```

! Access Switch Trunk Links(for public, admin, and new office floor)

```
interface range FastEthernet0/1 - 3  
switchport mode trunk  
switchport trunk native vlan 99
```

3. Admin Switch (SW2-Admin)

Show uplink trunk port and access ports assigned to VLAN 10:

! Uplink Trunk to Core Switch

```
interface FastEthernet0/24  
switchport mode trunk  
switchport trunk native vlan 99
```

! Admin End-Device(3 PC's for now) Access Ports

```
interface range FastEthernet0/1 - 3  
switchport mode access  
switchport access vlan 10
```

4. Public Switch (SW3-Public)

Show uplink trunk port and access port assigned to Public Wi-Fi (VLAN 20):

! Uplink Trunk to Core Switch

```
interface FastEthernet0/24
```

```
switchport mode trunk
switchport trunk native vlan 99
```

! Wireless Access Point Port

```
interface FastEthernet0/1
switchport mode access
switchport access vlan 20
```

5. Floor 2 Switch (SW4-Floor2-CR2)

Show uplink trunk port and access ports assigned to CR2 Floor expansion (VLAN 30):

! Uplink Trunk to Core Switch

```
interface FastEthernet0/24
switchport mode trunk
switchport trunk native vlan 99
```

! Floor 2 Wired & Wireless Access Ports

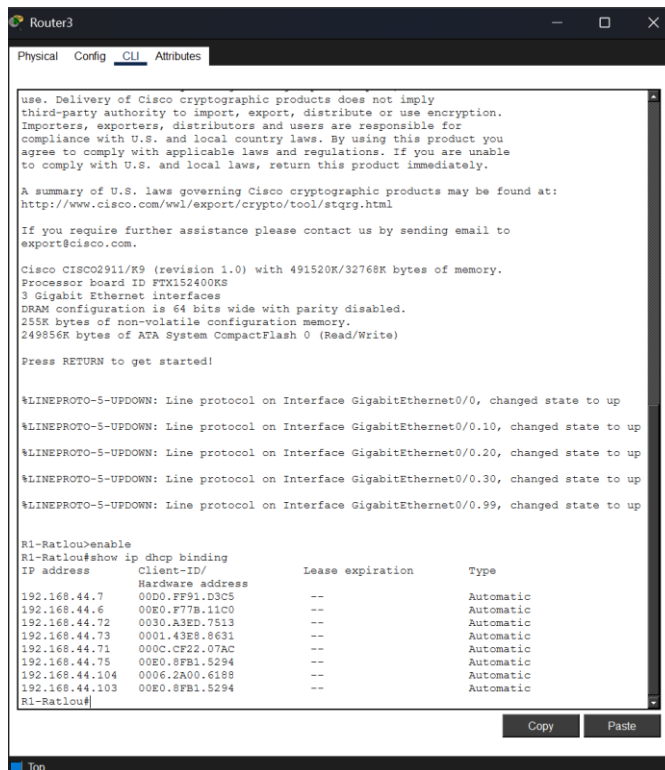
```
interface range FastEthernet0/1 - 2
switchport mode access
switchport access vlan 30
```

!

```
interface FastEthernet0/10
switchport mode access
switchport access vlan 30
```

7. Testing & Verification

- **DHCP Verification:** Output from **Show ip dhcp binding** on R1-Ratlou.



The screenshot shows the CLI of Router3. The tabs at the top are Physical, Config, CLI (selected), and Attributes. The output text is as follows:

```
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

R1-Ratlou>enable
R1-Ratlou#show ip dhcp binding
IP address      Client-ID/      Lease expiration        Type
                Hardware address
192.168.44.7     00D0.F9B1.D3C5   --                       Automatic
192.168.44.6     00E0.F77B.11C0   --                       Automatic
192.168.44.72    0030.A3ED.7513   --                       Automatic
192.168.44.73    0001.43E8.8631   --                       Automatic
192.168.44.71    000C.C922.07AC   --                       Automatic
192.168.44.75    00E0.8FB1.5294   --                       Automatic
192.168.44.104   0006.2A00.6188   --                       Automatic
192.168.44.103   00E0.8FB1.5294   --                       Automatic
R1-Ratlou#
```

Buttons for Copy and Paste are at the bottom right. A Top button is at the bottom left.

- **VLAN Database Verification:** Output from **show vlan brief** on SW1-Core.



The screenshot shows the CLI of SW1-Core. The tabs at the top are Physical, Config, CLI (selected), and Attributes. The output text is as follows:

```
SW1-Core con0 is now available

Press RETURN to get started.

SW1-Core>enable
SW1-Core#show vlan brief

VLAN Name      Status      Ports
-----
1  default      active      Fa0/4, Fa0/5, Fa0/6, Fa0/7
                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                Fa0/24, Gig0/2
10  Admin-Staff  active
20  Public-Area  active
30  CR2-New-Floor active
99  Management   active
1002 fddi-default  active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active
SW1-Core#
```

Buttons for Copy and Paste are at the bottom right. A Top button is at the bottom left.

- **Trunk Link Verification:** Output from **show interfaces trunk**.

```
SW1-Core>enable
SW1-Core#show vlan brief

VLAN Name                Status Ports
-----
1    default                active Fa0/4, Fa0/5, Fa0/6, Fa0/7,
    Fa0/8, Fa0/9, Fa0/10, Fa0/11,
    Fa0/12, Fa0/13, Fa0/14, Fa0/15,
    Fa0/16, Fa0/17, Fa0/18, Fa0/19,
    Fa0/20, Fa0/21, Fa0/22, Fa0/23,
    Fa0/24, Gig0/2

10   Admin-Staff             active
20   Public-Area             active
30   CR2-New-Floor           active
99   Management               active
1002 fddi-default            active
1003 token-ring-default     active
1004 fddinet-default        active
1005 trnet-default          active

SW1-Core#show interfaces trunk

Port      Mode      Encapsulation  Status  Native vlan
-----
Fa0/1     on        802.1q         trunking  99
Fa0/2     on        802.1q         trunking  99
Fa0/3     on        802.1q         trunking  99
Gig0/1    on        802.1q         trunking  99

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20,30,99
Fa0/2     1,10,20,30,99
Fa0/3     1,10,20,30,99
Gig0/1    1,10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20,30,99
Fa0/2     1,10,20,30,99
Fa0/3     1,10,20,30,99
--More--
```

- **End-to-End Ping Verification:** Successful ping output from Floor 2 Wireless Laptop to Gateway **192.168.44.1**:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.44.1

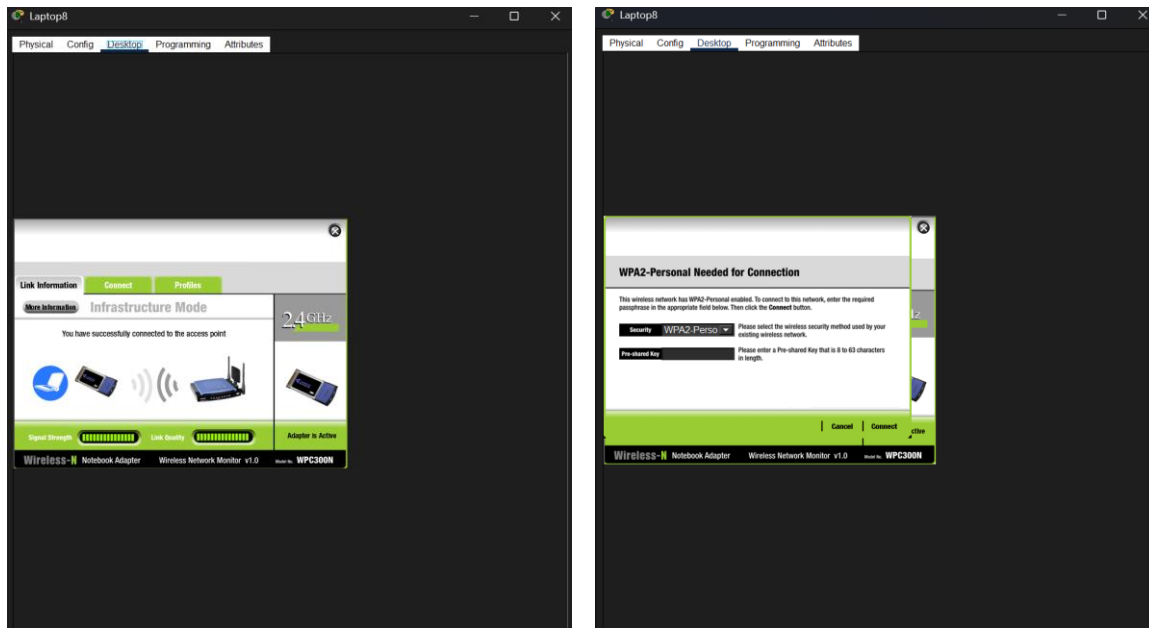
Pinging 192.168.44.1 with 32 bytes of data:

Reply from 192.168.44.1: bytes=32 time=24ms TTL=255
Reply from 192.168.44.1: bytes=32 time=23ms TTL=255
Reply from 192.168.44.1: bytes=32 time=15ms TTL=255
Reply from 192.168.44.1: bytes=32 time=20ms TTL=255

Ping statistics for 192.168.44.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 25ms, Average = 21ms

C:\>
```

- WPA2-PSK Authentication for Wireless Office floor laptop:



8. Future Milestones Plan

- **Milestone 2 (02 October 2026):** Full implementation of Change Request 2 (CR2 expansion area) and complete Fault Isolation Scenario (controlled network breakdown, diagnostic logs, and resolution proof).
- **Final Submission (16 October 2026):** Final tested .pkt file, complete GitHub repository documentation, and 15–20 minute screen-recorded video presentation featuring webcam overlay and live connectivity demonstration.

9. Preliminary Project Reflection

Developing the foundational network for the Ratlou Local Municipality highlighted the importance of structured VLSM planning before applying CLI configurations. Managing native VLAN matching across trunk links initially triggered Spanning Tree protocol alerts, but aligning native VLAN 99 across all switches resolved the issue. The configuration successfully achieves inter-VLAN routing while isolating public Wi-Fi traffic from administrative databases.